

Narendra Kumar Kotipalli

GenAI Engineer | Frontend Developer | React.js | Next.js | TypeScript | AI/Agent Interfaces | Python | FastAPI

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Portfolio: <https://narendra-kumar-kotipalli.vercel.app/>

Professional Summary

Frontend-focused GenAI Engineer with **3 years of experience** building scalable, high-performance web applications using **React, Next.js (App Router), and TypeScript**. Experienced in developing **reusable component architectures**, **optimizing Core Web Vitals**, and **building responsive, accessible UIs following WCAG 2.1 AA standards**. Strong expertise in **Redux Toolkit, Context API, REST and GraphQL API integration**, and **full-stack collaboration with Python/FastAPI backends**. Proven track record of improving application performance, reducing load times by 25%, decreasing production defects by 30%, and mentoring junior engineers.

Professional Skills

Front-End:

React.js (Hooks, Suspense, Server Components), React 18, Next.js (App Router, SSR, SSG, ISR), TypeScript, JavaScript (ES6+), Redux Toolkit, TanStack Query, Context API, Component Architecture, Performance Optimization, Design Patterns, Vite, Webpack, Custom Hooks

UI & Styling:

HTML5, CSS3, Tailwind CSS, Material UI, Bootstrap, SCSS, Web Accessibility (WCAG 2.1 AA), Semantic HTML, ARIA, Responsive Web Design, Cross-Browser Compatibility, Design Systems

Performance Optimization:

Core Web Vitals (LCP, INP, CLS), Lighthouse, Code Splitting, Lazy Loading, Memoization, React Rendering Optimization, Bundle Analysis, Chrome DevTools, React DevTools Profiler, Performance Auditing

Back-End (Full-Stack):

Python, FastAPI, RESTful APIs, REST API Development, JWT Authentication, Database Design

GenAI & AI Tools:

GitHub Copilot, Cursor, Builder.io, Claude, Prompt Engineering, AI Integration

Cloud, Databases & DevOps:

Docker, GCP, Cloud Storage, BigQuery, GitLab CI/CD, MongoDB, PostgreSQL, Redis, CI/CD Automation, GitHub Actions

Tools, Testing & Practices:

Git, Bitbucket, Postman, JIRA, Agile/Scrum, API Testing, Jest, Vitest, React Testing Library (RTL), Cypress, Playwright, Mock Service Worker (MSW), axe-core, Code Coverage Analysis, Environment Configuration (.env), Version Control Best Practices, Secure Coding Practices

Professional Experience

GenAI Engineer (Frontend Focus) - FIS Clouds Pvt. Ltd | Hyderabad (Feb 2024 – Jul 2026)

Gurita AI — Agent Automation & AI Workflow Platform

Tech Stack: React, TypeScript, Next.js, Redux Toolkit, React Flow, Python, FastAPI, MongoDB, PostgreSQL.

- Architected a scalable **React/Next.js frontend** for an AI agent automation platform, supporting complex workflow creation, configuration, execution, and monitoring across a microservices-based architecture.
- Built a **node-based workflow builder using React Flow and Redux Toolkit**, enabling users to visually create, connect, configure, validate, and manage multi-step AI automation workflows.
- Implemented **real-time workflow execution using WebSockets**, synchronizing node status, execution progress, logs, failures, and completion states without requiring page refreshes.
- Improved **application load time by 25%** through code splitting, lazy loading, rendering optimization, and bundle-level performance analysis.

- Developed reusable **component architecture and UI design-system components** for workflow nodes, forms, configuration panels, dialogs, dashboards, and common interaction patterns.
- Integrated **REST and GraphQL APIs** with client-side caching and asynchronous state management, reducing redundant API requests and improving responsiveness across data-intensive workflow screens.
- Implemented **internationalization (i18n) for 9 languages**, supporting localized UI content and language-aware application workflows for global users.
- Built responsive and accessible interfaces using **semantic HTML, ARIA, and WCAG 2.1 AA practices** across drag-and-drop workflows, forms, dialogs, and dashboards.

Curie — AI-Powered Research Platform

Tech Stack: React, TypeScript, MUI, Python, FastAPI, PostgreSQL.

- Built production-grade **React interfaces for AI-powered document generation, summarization, data extraction, and research workflows**, with responsive layouts and real-time result updates.
- Developed reusable **React component architecture and custom hooks** for document workflows, analysis views, research dashboards, and result-review interfaces.
- Integrated **React frontend with Python/FastAPI services** for AI processing, document parsing, structured data extraction, validation, and asynchronous result handling.
- Optimized frontend-backend data flows for **large document and research datasets**, improving request handling, state synchronization, and rendering efficiency.
- Built interactive **3D force-directed graph visualizations** for hypothesis mapping, entity relationships, and research connections, enabling visual exploration of complex datasets.
- Implemented structured **loading, error, empty, retry, and asynchronous processing states** across AI workflows to provide reliable user feedback during long-running operations.

RPA Web Automation Platform

Tech Stack: Python, FastAPI, React, Redux Toolkit, PostgreSQL, Redis.

- Built a **browser automation platform** using React and FastAPI for workflow recording, configuration, execution, and monitoring of multi-step automation processes.
- Designed workflow configuration and execution interfaces using **Redux Toolkit** to manage complex application state, execution status, validation, errors, and user actions.
- Implemented **Redis caching**, reducing workflow execution time by **30%** and improving responsiveness for frequently accessed workflow and execution data.
- Developed secure **credential and data-injection workflows** with encrypted sensitive configuration handling to protect automation-related data.
- Designed reusable automation workflows supporting execution across **multiple environments**, separating environment-specific configuration from core workflow definitions.
- Implemented **unit and integration testing using Jest and React Testing Library**, covering reusable components, workflow logic, and critical user interactions.

Web Developer Intern (6 months) - Visual Path | Hyderabad (Jul 2023 – Jan 2024)

- Developed responsive and accessible UI using React for real-time appointment booking and scheduling
- Integrated REST APIs for data fetching and state management, enabling frontend-backend communication
- Implemented real-time doctor availability and scheduling logic, improving user experience and booking efficiency
- Built reusable form components using HTML5, CSS3, and JavaScript, ensuring cross-browser compatibility and maintainability
- Automated email notification system and data handling workflows, reducing manual effort

Education

Aditya College of Engineering, Surampalem

Bachelor of Technology (B.Tech) in Mechanical Engineering

July 2018 – May 2021